SQL

SQL Interview Pack

8 patterns that cover most interview questions, plus 20 questions to test yourself.

Order of execution (asked in almost every round)

```
FROM → JOIN → WHERE → GROUP BY → HAVING → SELECT → DISTINCT → ORDER BY → LIMIT
```

This is why you cannot use a SELECT alias inside WHERE, but you can inside ORDER BY.

The 8 patterns

1. Top N per group

```
SELECT * FROM (
  SELECT e.*, DENSE_RANK() OVER (PARTITION BY dept ORDER BY salary DESC) rk
  FROM employees e) t
WHERE rk <= 3;
```

2. Running total

```
SELECT order_date, amount,
       SUM(amount) OVER (ORDER BY order_date) AS running_total
FROM orders;
```

3. Month over month change

```
SELECT month, revenue,
       revenue - LAG(revenue) OVER (ORDER BY month) AS mom_change
FROM monthly_revenue;
```

4. Remove duplicates, keep latest

```
DELETE FROM t WHERE id IN (
  SELECT id FROM (
    SELECT id, ROW_NUMBER() OVER (PARTITION BY email ORDER BY created_at DESC) rn
    FROM t) x WHERE rn > 1);
```

5. Consecutive days (gaps and islands)

```
SELECT user_id, MIN(login_date), MAX(login_date), COUNT(*) AS streak
FROM (
  SELECT user_id, login_date,
         login_date - ROW_NUMBER() OVER (PARTITION BY user_id ORDER BY login_date) * INTERVAL
'1 day' AS grp
  FROM logins) x
GROUP BY user_id, grp HAVING COUNT(*) >= 3;
```

6. Find what is missing (anti join)

```
SELECT c.* FROM customers c
LEFT JOIN orders o ON o.customer_id = c.id
WHERE o.id IS NULL;
```

7. Conditional aggregation (pivot without PIVOT)

```
SELECT region,
       SUM(CASE WHEN status='Delivered' THEN 1 ELSE 0 END) AS delivered,
       SUM(CASE WHEN status='Returned'  THEN 1 ELSE 0 END) AS returned
FROM orders GROUP BY region;
```

8. Self join (employee and manager)

```
SELECT e.name AS employee, m.name AS manager
FROM employees e LEFT JOIN employees m ON e.manager_id = m.id;
```

The JOIN trap: if your JOIN doubled the row count, the join key is not unique on one side. Check with `SELECT key, COUNT(*) FROM t GROUP BY key HAVING COUNT(*) > 1`. Fix by deduplicating first or aggregating before the join.

RANK vs DENSE_RANK vs ROW_NUMBER

Salary	ROW_NUMBER	RANK	DENSE_RANK
900	1	1	1
900	2	1	1
800	3	3	2

20 questions to test yourself

1. Second highest salary without LIMIT/TOP.
2. Difference between WHERE and HAVING.
3. Customers who ordered in Jan but not Feb.
4. Top 3 products by revenue in each category.
5. 7-day rolling average of sales.
6. Users who logged in 3 days in a row.
7. Find duplicate emails and keep the latest row.
8. Percentage of total revenue per region.
9. Employees earning more than their manager.
10. First order date of every customer.
11. Month on month growth %.
12. Difference between UNION and UNION ALL.
13. INNER vs LEFT JOIN output when keys repeat.
14. COUNT(*) vs COUNT(col) vs COUNT(DISTINCT col).
15. Why does NULL = NULL not return true?
16. CTE vs subquery vs temp table.
17. Retention: users active in month 1 who came back in month 2.
18. Median salary per department.

19. Cumulative distinct users by day.
20. How you would speed up a slow query.

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